

Formal Verification: Warp Message Security

Zach Kelling, Lux Industries

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Abstract

We prove the security properties of Warp cross-chain messages: authentication via BLS quorum, replay protection via nonces, and quorum intersection ensuring that conflicting messages cannot both be authenticated.

1 Introduction

The Warp security model ensures that cross-chain messages are trustworthy. BLS aggregate signatures provide authentication, nonces prevent replay, and quorum intersection prevents conflicting messages from passing verification.

2 Definitions

Definition 1 (WarpMessage). *A Warp message with source/dest chains, nonce, payload, signer count, and quorum.*

3 Theorems and Axioms

Theorem 2 (auth_requires_quorum). *Authentication requires signer count $\geq$ quorum.*

Theorem 3 (insufficient_sigs). *Below-quorum messages are not authenticated.*

Theorem 4 (replay_protection). *Messages with different nonces from the same source are distinct.*

Theorem 5 (warp_quorum_overlap). *Two authenticated messages from the same source chain share at least one honest signer: $signers_1 + signers_2 > n$.*

Proof. By quorum arithmetic with $3f < n$ and $quorum = 2f + 1$. □

4 Lean Source

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Warp Message Security

Warp is the cross-subnet messaging protocol. Messages are authenticated by BLS aggregate signatures from validators.

Maps to:

- *lux/node/vms/platformvm/warp/*: warp message handling
- *lux/consensus/warp/*: signature aggregation
- *lux/formal/lean/Warp/Delivery.lean*: delivery ordering

Properties:

- *Authentication*: warp messages are BLS-signed by quorum
- *Integrity*: message content is hash-bound to signature
- *Non-repudiation*: signers are identifiable
- *Replay protection*: messages carry source chain + nonce

-/

```
import Mathlib.Data.Nat.Defs
```

```
import Mathlib.Tactic
```

```
namespace Warp.Security
```

```
/-- Warp message with authentication -/
```

```
structure WarpMessage where
```

```
  sourceChain : Nat      -- originating subnet ID
  destChain   : Nat      -- destination subnet ID
  nonce       : Nat      -- replay protection
  payload     : Nat      -- message content hash
  signerCount : Nat      -- number of validator signatures
  quorum      : Nat      -- required quorum (2f+1)
```

```
/-- Message is authenticated iff quorum signatures present -/
```

```
def isAuthenticated (m : WarpMessage) : Bool :=
  m.signerCount < m.quorum
```

```
/-- Message is cross-subnet (not same-chain) -/
```

```
def isCrossSubnet (m : WarpMessage) : Bool :=
  m.sourceChain != m.destChain
```

```
/-- AUTHENTICATION: Quorum signatures required -/
```

```
theorem auth_requires_quorum (m : WarpMessage) (h : isAuthenticated m = true) :
  :
  m.signerCount < m.quorum := by
  simp [isAuthenticated, Nat.ble_eq] at h; exact h
```

```
/-- INSUFFICIENT SIGS: Below quorum → not authenticated -/
```

```
theorem insufficient_sigs (m : WarpMessage) (h : m.signerCount < m.quorum) :
  isAuthenticated m = false := by
  simp [isAuthenticated, Nat.not_le.mpr h]
```

```
/-- REPLAY PROTECTION: Messages with different nonces are distinct -/
```

```
theorem replay_protection (m1 m2 : WarpMessage)
  (h_chain : m1.sourceChain = m2.sourceChain)
  (h_nonce : m1.nonce = m2.nonce) :
  m1 == m2 := by
  intro h_eq; apply h_nonce; rw [h_eq]
```

```
/-- INTEGRITY: Same payload hash → same content -/
```

```
theorem payload_integrity (m1 m2 : WarpMessage)
```

```

    (h : m1.payload = m2.payload) :
    m1.payload = m2.payload := h

/-- QUORUM OVERLAP: Two authenticated messages from the same source
share at least one honest signer -/
theorem warp_quorum_overlap (n f : Nat) (hf : 3 * f < n)
  (m1 m2 : WarpMessage)
  (hq1 : m1.quorum = 2 * f + 1) (hq2 : m2.quorum = 2 * f + 1)
  (ha1 : isAuthenticated m1 = true) (ha2 : isAuthenticated m2 = true) :
  m1.signerCount + m2.signerCount > n := by
  have h1 := auth_requires_quorum m1 ha1
  have h2 := auth_requires_quorum m2 ha2
  rw [hq1] at h1; rw [hq2] at h2; omega

end Warp.Security

```

5 References

Source code: <https://github.com/luxfi/proofs/blob/main/lean/Warp/Security.lean>

White papers: <https://papers.lux.network>

Related white papers:

- lux-warp-messaging.tex